

with potentially work-related low back symptoms. Findings of the medical history and physical examination may alert the examiner to other pathology (e.g., not of low back origin) that can present as low back disorders. In this assessment, certain findings, referred to as red flags, raise suspicion of serious underlying medical conditions (see Table 1). The absence of red flags and conditions rules out the need for special studies, referral, or inpatient care during the first 4 to 6 weeks. During this time, spontaneous recovery is expected, provided any associated workplace factors are mitigated.(29)

There also are psychological red flags that should be evaluated, such as PTSD, suicidality, hallucinations or intoxication, which have been called primary risk factors,(49) and have been reviewed elsewhere.(50) Suicidality though is a potentially fatal complication, which makes it a more severe complication than cauda equina.

Red Flags

Potentially serious disorders are referred to as “red flags.” These include acute fractures, acute dislocations infection, tumor, progressive neurologic deficit, or cauda equina syndrome.

Table 1. Red Flags for Potentially Serious Low Back Conditions

Disorder	Medical History	Physical Examination/Diagnostic Testing
SPINAL DISORDERS		
Fracture	Major trauma, such as vehicular accident or fall from height Minor trauma or supra-maximal lifting in older or potentially osteoporotic patients	Percussion tenderness over specific spinous processes Careful neurological examination for signs of neurological compromise
Tumor and Neoplasia	Severe localized pain over specific spinal processes History of cancer Age >50 years Constitutional symptoms, such as recent unexplained weight loss or fatigue Pain that worsens when patient is supine Pain at night or at rest	Pallor, reduced blood pressure, diffuse weakness Tenderness over spinous process and percussion tenderness Decreased range of motion due to protective muscle spasm History of sciatica for detection of cancer [†] ▪ Sciatica sensitivity = 58 to 93% ▪ Sciatica specificity = 78% History of paresthesia for detection of cancer [†] ▪ Paresthesia sensitivity = 58% Plain radiography for detection of cancer [‡] ▪ Radiography sensitivity = 60% ▪ Radiography specificity = 90 to 99.5% Magnetic resonance imaging (MRI) for detection of cancer [‡] ▪ MRI sensitivity = 83 to 93% ▪ MRI specificity = 90 to 97% Radionuclide scanning for detection of cancer [‡] ▪ Planer imaging sensitivity = 74 to 98% ▪ Planer imaging specificity = 64 to 81%